

Introduction to Data mining

What is data mining? Examples.

Data mining is the process of searching and analyzing a large batch of raw data in order to identify patterns and extract useful information.

Data mining is used to explore large data volumes to find patterns and insights that can be used for specific purposes. These purposes might include improving sales and marketing, optimizing manufacturing, detecting fraud, and enhancing security.

Why data mining?

Data mining is important because it helps organizations and individuals extract useful insights from large amounts of data.

Enhances Business Strategies

- Companies use it for customer segmentation and personalized marketing.
- Helps in inventory management, pricing strategies, and customer retention.

Improves Decision-Making

- Businesses can make data-driven decisions instead of relying on intuition.
- Governments and healthcare industries use it to improve public services.

Medical and Scientific Discoveries

- Used in disease prediction, drug discovery, and patient care improvements.
- Helps in genome research and bioinformatics.

Fraud Detection and Risk Management

- Banks and financial institutions use data mining to detect unusual transactions.
- Helps in identifying cybersecurity threats.

Discover Patterns and Trends

- Helps identify hidden patterns and correlations in data.
- Predicts customer behavior, fraud detection, and market trends.

Automation and AI Development

- Supports machine learning models by providing meaningful data patterns.
- Used in recommendation systems like Netflix and Amazon.

Competitive Advantage

- Helps businesses stay ahead of competitors by understanding market trends.
- Assists in predicting future sales and customer needs.

What is (not) Data Mining?

What is not Data Mining?

- Look up phone number in phone directory
- Query a Web search engine for information about “Amazon”

What is Data Mining?

-Certain names are more prevalent in certain US locations (O’Brien, O’Rourke, O’Reilly... in Boston area)

- Group together similar documents returned by search engine according to their context (e.g. Amazon rainforest, Amazon.com,)
- SPL -> DSA-I

Origins of Data Mining:

- Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems
- Traditional Techniques may be unsuitable due to
 - Enormity | (extreme scale) of data
 - High dimensionality (large number of features or attributes) of data
 - Heterogeneous, distributed nature of data

Related technologies:

- Machine learning: Machine learning (ML) is a type of artificial intelligence (AI) that allows computers to learn and improve from data without being explicitly programmed. It uses algorithms to analyze data, identify patterns, and make decisions.
- OLAP : Online analytical processing (OLAP) is a technology that analyzes large amounts of data quickly. It's used in business intelligence, decision support, and reporting.
- Statistics: Statistics is a branch of applied mathematics that involves the collection, description, analysis, and inference of conclusions from quantitative data.
- DBMS: A database management system (DBMS) is **a software system for creating and managing databases**. A DBMS enables end users to create, protect, read, update, and delete data in a database. It also manages security, data integrity, and concurrency for databases.

Data Mining Tasks

Prediction Methods- Use some variables to predict unknown or future values of other variables

Description Methods- Find human-interpretable patterns that describe the data

Classification [Predictive]

-Classification involves assigning data into predefined categories based on specific attributes. For example, using algorithms trained on labeled data, emails can be classified as 'spam' or 'not spam'.

-Clustering groups data into clusters based on similarities without predefined labels.

Regression [Predictive]- Predict a value of a given continuous valued variable based on the values of other variables

Examples:

- Predicting sales amounts of new product based on advertising expenditure.
- Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
- Time series prediction of stock market indices

Linear-> age and height

Nonlinear -> population growth with time

Logistics -> diabetes based on some issues

Deviation Detection [Predictive]

-Abnormality instead of normal occurrence [credit card fraud, Network Intrusion Detection]

Clustering [Descriptive]

Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that

- Data points in one cluster are more similar to one another.
- Data points in separate clusters are less similar to one another.

Association Rule Discovery [Descriptive]

Rules which will predict occurrence of an item based on occurrences of other items.

- *Customers who buy bread are likely to also buy milk.*

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

Sequential Pattern Discovery [Descriptive]

Find rules that predict strong sequential dependencies among different events. Event occurrences in the patterns are governed by timing constraints.

- *People who buy DVD players tend to buy DVD in the period immediately following the purchase*
- *Candy sales peak before Halloween (timing constraints)*
- *Athletic Apparel Store:*
(Shoes) (Racket, Racketball) --> (Sports_Jacket)

What is Machine Learning?

Machine learning (ML) is a type of artificial intelligence (AI) that allows machines to learn and improve from experience.

Types of relationship

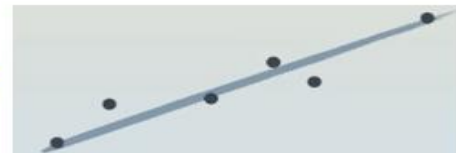
-Nonlinear relationship

-Linear relationship

□ Types of relationships between x and t

o Linear relationship:

- $t = w_0 + w_1x$ (w_0, w_1 : model parameters)



o Non-linear relationship:

- $t = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=1}^M w_jx^j$



o Model parameters are something that **need to be determined** somehow.

Math for coefficients and least square equation for age salary

Straight-line regression using the method of least squares. Table 1 shows a set of paired data where x is the number of works experience of a college graduate and y is the corresponding salary of the graduate. Estimate the equation of least squares line. Also, calculate the salary y for 10 years experiences.

Salary data.

x years experience	y salary (in \$1000s)
3	30
8	57
9	64
13	72
3	36
6	43
11	59
21	90
1	20
16	83

Solution:

Mean value of x , $\bar{x} = 9.1$ and Mean value of y , $\bar{y} = 55.4$

We know that,

$$w_1 = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|} (x_i - \bar{x})^2}$$

$$w_1 = \frac{(3 - 9.1)(30 - 55.4) + (8 - 9.1)(57 - 55.4) + \dots + (16 - 9.1)(83 - 55.4)}{(3 - 9.1)^2 + (8 - 9.1)^2 + \dots + (16 - 9.1)^2} = 3.5$$

Also we know,

$$\bar{y} = w_1 \bar{x} + w_0$$

$$w_0 = 55.4 - (3.5)(9.1) = 23.6$$

$$\text{Now, } y = w_0 + w_1 x \Rightarrow 23.6 + 3.5x \Rightarrow 23.6 + 3.5 \cdot 10 = 58.6 \cdot 1000 = \$58600$$

ML vs Data mining

Data mining is used on an existing dataset (like a data warehouse) to find patterns. Machine learning, on the other hand, is trained on a 'training' data set, which teaches the computer how to make sense of data, and then to make predictions about new data sets.

Data Mining	Machine Learning
Extracting useful information from large amount of data	Introduce algorithm from data as well as from past experience
Used to understand the data flow	Teaches the computer to learn and understand from the data flow
Huge databases with unstructured data	Existing data as well as algorithms
Data mining abstract from the data warehouse	Machine learning reads machine
Clustering, association rule mining, outlier detection	Regression, classification, clustering, deep learning
Strong domain knowledge is often required	Domain knowledge is helpful, but not always necessary

Stages of the Data mining Process:

Four main stages:

1. **Data Pre-processing:** Data preprocessing is the process of preparing data for machine learning and other data analysis. It involves:

- **Data cleaning:** Fixing errors, removing duplicates, and handling missing values
- **Data transformation:** Scaling, normalizing, and encoding categorical variables
- **Data reduction:** Selecting relevant features and reducing dimensionality
- **Data integration:** Combining data from multiple sources
- **Data formatting:** Ensuring consistent data types and structures
- **Data validation:** Checking for errors, ensuring data consistency, and verifying that transformations were applied correctly

2. Exploratory Data Analysis: Exploratory Data Analysis (EDA) is an analysis approach that identifies general patterns in the data. These patterns include outliers and features of the data that might be unexpected. EDA is an important first step in any data analysis.

3. Data Selection: Data selection is the process of choosing the right data type, source, and collection instruments for a project. It's a crucial step before data collection.

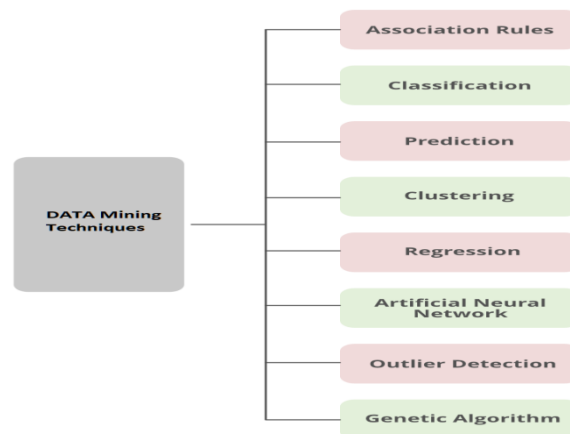
Purpose

- To ensure that the data is relevant, accurate, and aligned with the project's goals
- To answer research questions
- To train or evaluate a machine learning model

4. Knowledge Discovery: Knowledge discovery is the process of extracting useful knowledge from data

Data Mining Techniques:

- Classification
- Clustering Regression
- Associative rules
- Sequential Pattern
- Artificial Neural Network
- Outlier detection
- Prediction
- Genetic Algorithm



Prediction is a data mining technique that involves using models to predict future outcomes. This is called predictive data mining.

Genetic algorithms (GAs) are a data mining technique that can be used to classify data and solve optimization problems. GAs are based on natural evolution and can be used to find optimal solutions.